

I work on interactive and learning-based algorithms for simulating crowds and modeling the environments they move through. My contributions fall in two connected areas: simulation of multi-agent motion, from pedestrians to ant swarms^{1,2,3,4,5,6,7,8,9,10,11,12,13,14}, and the analysis and synthesis of 3D scenes, which supply the spaces those agents inhabit^{15,16,17,18,19,20,21,22,23,24,25,26}. These areas are converging on a single goal: realistic, data-grounded virtual environments in which populations of simulated agents can be studied at scale. Such environments allow planners, designers, engineers, and researchers to explore how people move through a space and how changes to it affect outcomes.

My research has been recognized by the *ACM SIGGRAPH Conference on Motion in Games*, where I received the **Best Paper Award**¹⁰, by the *ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games*, where I received the **Best Student Paper Award**, and by the *ACM SIGGRAPH/Eurographics Symposium on Computer Animation*, where I received the **Best Presentation Award**¹⁴. In 2018, I was also an internationally selected **finalist** at both the *ACM SIGGRAPH Thesis Fast Forward* and the *ACM SIGGRAPH Asia Doctoral Consortium*.

My future research agenda aims to advance interdisciplinary approaches at the intersection of artificial intelligence, and human-centered computing. I am particularly interested in combining machine learning and physically grounded simulation to study collective decision-making, social interaction, and emergent behavior in both virtual and real-world settings. I am also interested in exploring how generative AI and reinforcement learning can support intelligent tutoring systems, virtual training environments, and assistive technologies that adapt to human behavior in real time.

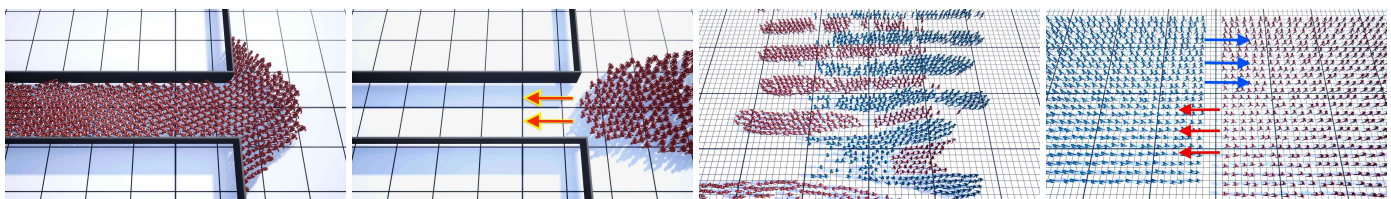
The two thrusts below summarize this work. Thrust I develops crowd dynamics that run in real time at 100,000+ agents and learn behavior from data, and Thrust II develops scene synthesis and understanding systems.

THRUST I: CROWDS, GROUPS, AND MULTI-AGENT MOTION DYNAMICS

Multi-agent simulation has applications in entertainment, pedestrian analysis, urban planning, robotics, and autonomous systems. A **crowd** is a collection of independent, self-actuated agents. Each agent has individual navigational goals in this shared environment. Since agents share their environment, they can interact and collide with each other. Agent movement is controlled by a navigation algorithm, which must ensure that **agents progress towards their goals, while avoiding collisions**. However, computing collision-free agent motion is difficult due to the complexity of such dynamic interactions. Field experiments are one important avenue for testing a navigation algorithm. A more efficient and less costly approach is to conduct virtual simulation experiments, which allow quick insight into the dynamics and performance of such navigation algorithms. Furthermore, a simulation allows us to easily test “what-if” scenarios.



Agent

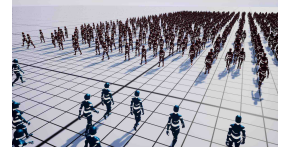


(a) A group of agents passing through a narrow corridor.

(b) Two groups exchanging positions.

Figure 1: Emergent phenomena in crowd simulation. Our results¹⁰ recreate different phenomena observed in pedestrian crowds, including: (a) clogging and arching near bottlenecks²⁷, (b) groups self-organizing into lanes²⁸, with stable and real-time performance for +100K individuals. These results are made possible by framing the crowd motion as a constraint optimization problem, which is then efficiently solved using a GPU.

During a simulation, agents run through a continuous cycle of sensing and acting, where each cycle correlates to a time step. At the beginning of each cycle, each agent independently computes a trajectory to its goal, while avoiding collisions with other agents or obstacles. In addition to collision-free movement, a simulation should **capture both individual and group behavior** observed in real crowds, while being computationally interactive.



Multi-Agent Crowds

Physics-based Crowd Simulation

One core problem in multi-agent dynamics is to simulate dense, large crowds interactively. We proposed a method that simulates +100K agents in real time¹⁰, which we achieved using GPU parallelization, and a specialized physics-based solver that uses geometric constraints that capture agent-agent dynamics in a crowd (Fig. 1). **This result is currently the state-of-the-art for achieving fast crowd simulation.**

Later, we extended this framework to simulate groups moving in formations⁵, including **path planing** that allows groups of agents to maintain their formation while navigating⁶. We also developed methods for simulating non disc-shaped agents, since a human or robot figures may be better approximated with other geometry⁴, which allows capturing angular motion and turns more accurately. In addition, my group has also researched how other physics solver frameworks can be reformulated to simulate crowds²⁹. Such frameworks allow quicker computation for specific crowd simulation scenarios compared to prior work.

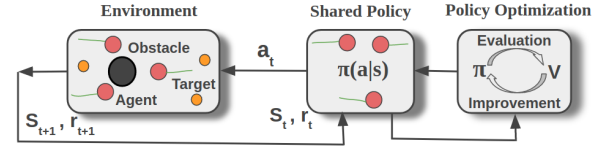


Figure 2: RL for multi-agent navigation¹².

Learning Crowd Behaviors

Physics solvers rules often fail to capture the somewhat stochastic nature of crowd dynamics. To attack this challenge, we proposed to learn how should crowd move using crowd sourcing via **Bayesian optimization**¹². Our method uses **deep reinforcement learning** for simulating crowds, where crowdsourcing is used to select policy hyper-parameters (Figure 2). Training agents with such parameters results in a crowd simulation that is preferred to users compared to state-of-the-art work.

Ongoing and future research directions

When simulating a crowd, the simulation is often only one part of the computational bottleneck, while visualizing an animated crowd is another. We proposed a **Perceptual Foveated Rendering** approach, where instead of rendering each agent with the same fidelity, we render agents that a viewer focuses on with a higher resolution, and crowds not in the area of Fovea with lower resolution (Figure 3). This is done by modifying the animation refresh rate³⁰.

Outside of pedestrian crowds, we also simulated 3D crowds⁸, including robot-like agents and ant swarm aggregations^{14,31}. Understanding the dynamics of ant swarms and bacteria colonies is important for many applications from health, traffic to manufacturing. This work modeled ants as self-actuating PBD particles, which spontaneous form elastic links and also maintain fluid-like properties such as maintaining a certain density within an aggregation (Figure 3).

We are extending our multi-agent methodology to bacterial motion, robotic agents, and other biological entities, with learning-based approaches to assist in creating realistic dynamics. We are also developing digital twins of real environments from street imagery, integrating computer vision, pedestrian simulation, traffic simulation, and behavioral modeling to produce realistic virtual scenes populated with pedestrians and vehicles. The framework will support planners, first responders, and researchers in exploring how emergency situations unfold and how interventions affect outcomes, validated against real-world observations. These topics are integrated in pending federal grant proposals.



Figure 3: Recent multi-agent research of our lab includes simulating crowds as capsule-shaped particles which more accurately capture the pedestrian profile than discs, simulating agents moving in formation (left), accelerating rendering of crowd simulations using foveated rendering (middle), and simulating non-human, 3D crowds such as ants (right).

THRUST II: SCENE UNDERSTANDING

Virtual worlds are challenging to construct, requiring professionals with extensive training in 3D modeling software. My lab’s research accelerates 3D content creation with computational methods for creating and understanding scenes.

One method we proposed uses a user-defined set of 3D objects and spatial arrangement constraints from which our method computes layout proposals (Figure 4). This is achieved by viewing layout synthesis through the lens of deformable body simulation. Both layouts and deformable bodies can be described by geometric constraints, which should be satisfied for a layout to be realistic, or for the movement of a body to be plausible. Both cases can be tackled using continuous optimization procedures. A layout synthesis solution is then an arrangement that satisfies such constraints³². The **novelty of our work**²¹ is in enabling **real-time, interactive synthesis of layouts**, even for **large scale** scenes, which **were previously intractable** via traditional optimization-based approaches³³. This work was later used to create a dataset of 3D spaces, which is **widely used today** for multiple deep learning tasks³⁴.

We also developed a **deep-learning generative transformer** approach for scene synthesis¹⁸, by formulating the synthesis task to be similar to text sequence generation. An interior is represented by a series of tokens, where the goal of the neural network is to generate a sequence of tokens that represent a realistic space. Our main innovation contrary to other transformer-based scene synthesis (and data-driven) work is that our synthesis **improves on the quality of rooms available in the original dataset**. This is done by using additional terms in the loss function, which encode ergonomic and geometric common-sense principles, such as the a table should be within reach of a person sitting.

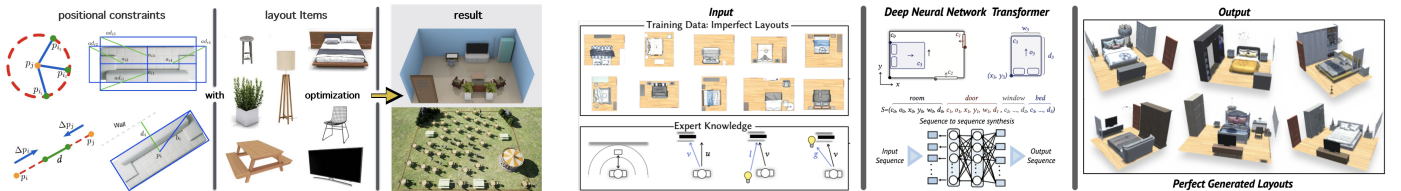


Figure 4: **Left:** Our results²¹ in 3D scene synthesis are faster by an order of magnitude than previous probabilistic methods and allow **real-time, interactive synthesis of scenes that were previously intractable**. Given input positional constraints and selected layout items, the algorithm rapidly outputs a variety of synthesized layouts. **Right:** We proposed a data-driven method for layout synthesis that combines interior expert knowledge with a data-driven generator based on a Transformer architecture¹⁸. The synthesized layouts integrate desirable properties not present in the original dataset.

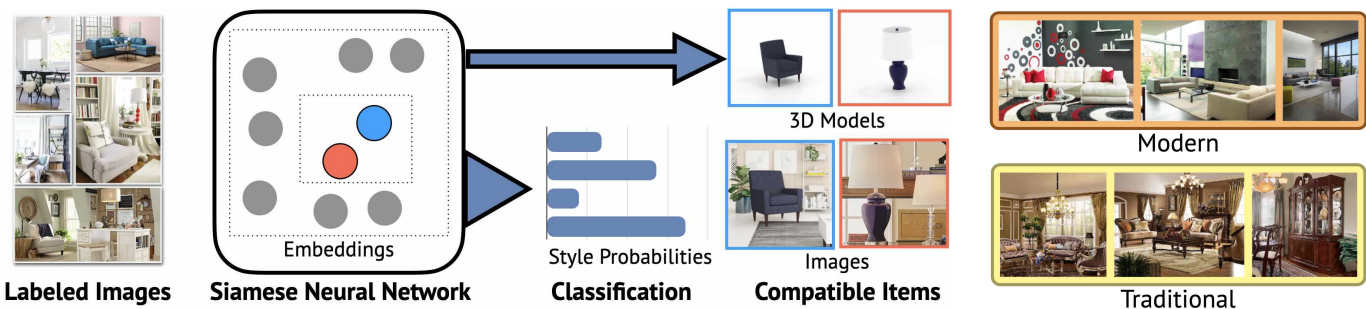
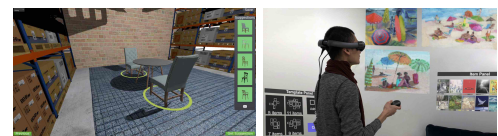


Figure 5: Left: Given style-labeled images of 3D objects and rooms, we trained a Siamese network to predict image styles.²⁴ After the network is trained, the stylistic compatibility between 3D objects is measured by selecting the nearest neighbors within the network’s embeddings for each image displaying the 3D object. Right: Examples of style labels. Our work can predict the predominant style of a space, which is useful for content creation, design, and retail¹⁵.

Ongoing and future research directions

Objects may not always fit a space, regardless of their function, due to differing visual features. For example, how well would a wheel fit into a car design? To understand this question better, we proposed an image-based classification approach for estimating 3D object and room style^{15,24}.

To accomplish this, we used (i) images where each object appears in multiple “styled” backgrounds (Figure 5), and (ii) image annotation labels that encode the style spectrum of each object in a relative manner. For example, a label indicates whether one table is more modern than another table. We have extended our style based work to Virtual Reality, and built interactive systems for quick synthesis of furniture in interior rooms based on interior design rules^{17,16}.



System for designing style-compatible space^{17,24}

We are currently investigating how clutter and a room’s organization affect the perception of space²⁵, as well as how suitable the room is for certain work tasks²⁶. This work, done jointly with perception researchers, compares computational clutter measures against human judgments, so that synthesized scenes can be evaluated by how people actually experience a space and not only by geometric plausibility. Our goal is to bring these perceptual measures back into layout synthesis, so that generated rooms are not only realistic but also comfortable and usable for the people and agents that occupy them.

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